

Consider a problem that is stated as follows. Three girls and three boys are on one side of a river, along with a boat that can hold one or two people. Find a way to get everyone to the other side without ever leaving a group of girls in one place outnumbered by the boys in that place. Formulate the problem with appropriate search space and do as per following instructions.

1. Implement and solve the problem optimally using the following search problem.
 - a) Depth Limited Search (take limit=3)
 - b) Iterative Deepening Search
2. Give comparison of both algorithms with the number of explored states and state approximate time complexities.